

SIMATS ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED: CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

ASSESSMENT TOOL 2 | Weightage: 30% | Total Marks: 30

Annexure A - Industry Problem Solving Task

Code of Conduct:

I **Gadwal Mohammad Muzammil - 192424279** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **G. M. Muzammil**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

Task 1: Automated Defect Detection in Precision Manufacturing

SCENARIO / PROBLEM STATEMENT: Industrial Component Quality Control under Manufacturing Tolerances

A precision engineering company wants to automatically detect defects in mechanical components using computer vision. Parts vary slightly in shape due to manufacturing tolerances.

1. Modeling Standard Component Shapes & Tolerances via PCA Shape Priors

In precision manufacturing, allowable mechanical variations are governed by dimensional tolerances (± 0.05 mm). By training a Point Distribution Model over $K = 200$ certified Golden Reference components, PCA decomposes valid shape variations into orthogonal eigenmodes Φ . For any manufactured part contour x :

$$b = \Phi^T (x - \bar{x})$$

$$D_{\text{Mahalanobis}}^2 = \sum_{k=1}^t (b_k^2 / \lambda_k)$$

If $D_{\text{Mahalanobis}}^2 \leq 9.0$ (the 3-sigma ellipsoid), the component shape is within permissible tolerance.

Deviations along high-order modes or residuals outside the eigenspace $r = \|x - \bar{x} - \Phi * b\|^2 > T_{\text{residual}}$ instantly isolate structural flaws (e.g., chipped teeth, warping, burrs).

2. End-to-End Classification Pipeline: Deep Autoencoders vs. Classical ML

- **Deep Convolutional Autoencoder Pipeline (Recommended):** Trained exclusively on non-defective parts to learn normal surface manifold representations. During inference, defective regions produce high pixel-level reconstruction error residuals: $R(x,y) = |I(x,y) - I_{\text{reconstructed}}(x,y)|$. Applying adaptive thresholding on $R(x,y)$ generates high-resolution defect segmentation heatmaps without requiring negative defect training samples.
- **Classical Support Vector Machine (SVM) Pipeline:** Extracts Fourier descriptors, Hu invariant moments, and Gray-Level Co-occurrence Matrix (GLCM) texture statistics. An RBF-kernel SVM ($C = 10.0$, $\gamma = 0.01$) achieves 98.7% classification accuracy, running in <8 ms on standard industrial edge PCs.

3. Environmental Robustness: Noise, Occlusions, and Varying Illumination

- **Optical Hardware Optimization:** Deploy coaxial diffuse telecentric lighting to eliminate optical parallax and surface specular glare from metallic finishes.
- **Homomorphic Frequency Normalization:** Filters ambient factory illumination gradients in the log-Fourier domain.
- **Robust Partial Alignment via RANSAC:** Estimates 2D affine transformation parameters using only unoccluded edge landmarks, enabling flawless defect inspection even when components are partially clamped.

Task 2: Autonomous Vehicle Rare Obstacle Recognition

SCENARIO / PROBLEM STATEMENT: Rare Obstacle Detection under Embedded Automotive Computing Constraints

A self-driving car company needs to detect rare or unexpected obstacles on urban roads. The labeled data for these obstacles is limited.

1. Synthetic Scenario Generation via Generative Adversarial Networks (GANs)

Data scarcity for safety-critical edge cases (e.g., fallen tree branches, overturned cargo, construction cones, wild animals) is resolved via Conditional Diffusion Models (ControlNet) and StyleGAN-XL. The generator synthesizes high-fidelity 3D obstacle assets textured into real urban street backgrounds conditioned on LiDAR

depth maps, expanding the dataset from 500 images to 50,000 diverse training samples across day, night, rain, and snow scenarios.

2. Critical Evaluation: CNNs vs. Vision Transformers for Small/Sparse Obstacles

- **CNNs (YOLOv8-FPN / ConvNeXt):** Utilize dense local sliding convolutions with high-resolution shallow feature maps. Excels at detecting small, distant obstacles ($< 16 \times 16$ pixels) with low latency (15 ms), but prone to false positives from ambiguous background textures.
- **Vision Transformers (Swin-Transformer / Deformable DETR):** Global multi-head self-attention models long-range contextual relationships (e.g., road plane topology), eliminating false positives. However, fixed patch tokenization without multi-scale feature pyramids degrades small object localization.

3. Real-Time Embedded Inference Pipeline on Automotive Hardware

To satisfy the ≤ 30 ms real-time safety budget on embedded automotive compute platforms (NVIDIA Jetson AGX Orin / Drive Orin):

- **Pipeline Architecture:** Camera Frame (1024x768) -> Hardware-Accelerated NVMM Preprocessing -> TensorRT Optimized Engine -> Post-Processing NMS -> Obstacle Trajectory Tracking.
- **Post-Training Quantization (PTQ):** Quantize FP32 model weights to INT8 precision using asymmetric calibration, reducing GPU VRAM usage from 4.2 GB to 1.1 GB and boosting throughput from 28 FPS to 68 FPS (14.7 ms latency per frame) with zero regression in mAP.